

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of M Sc Biotechnology/Microbiology Examination March 2018

Course code & Name: MS801 Genetic Engineering

Date: 26/03/2018 Day: Monday Time: 01:30 p.m To 04:30 p.m. Maximum Marks: 20
MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q - I Choose the correct answer for the following questions.

20

1. Which of the following is NOT the usual criterion for classification of plasmids?
(i) double stranded and single stranded (ii) Relaxed and Stringent
(iii) Conjugative and Non-conjugative (iv) None of the above
2. Which of the following features is NOT present in pBR322?
(i) *E. coli* origin of replication (ii) Unique HindIII site
(iii) Double antibiotic resistance genes for efficient selection (iv) Alpha complementation
3. Which of the following DNA modifying enzymes can add poly-dNTP tail at 3'OH end of double stranded DNA?
(i) Terminal transferase (ii) Taq DNA polymerase
(iii) Exonuclease (iv) PolyA kinase

4. Lambda phages with which of the following genotype are totally dependent upon recombination-mediated exchange for plaque formation?
 - (i) Red-Gam- (ii) Spi-
 - (iii) Rec+ Gam- (iv) Rec+/Red-
5. Which of the following is useful to convert blunt ends of double stranded DNA into cohesive ends?
 - (i) Adapters (ii) Linkers
 - (iii) Both of the above (iv) None of the above
6. Lambda phage progeny derived from C strain of *E. coli* lysogen cannot infect K-12 strain of *E. coli*. This phenotype is known as
 - (i) Restriction (ii) Lysogeny
 - (iii) Modification (iv) Spi⁺
7. Which of the following enzymes possesses a dual ability of a ligase and endonuclease?
 - (i) T4 DNA ligase (ii) DNA gyrase
 - (iii) Klenow fragment (iv) Topoisomerase
8. All *E. coli* plasmids replicate by _____.
 - (i) rolling circle mechanism (ii) theta mode of replication
 - (iii) iteron mediated mechanism (iv) None of the above
9. Vectors combining elements from both plasmids and M13 ori region are known as _____.
 - (i) Phasmids (ii) Phagemids
 - (iii) Cosmids (iv) BACs
10. *E. coli* is a less suitable host for protein expression because
 - (i) it lacks post translational modifications (ii) It forms protein aggregates
 - (iii) It may display codon usage bias (iv) All of the above
11. Which of the following is the disadvantage of cDNA library generated by hairpin loop strategy?
 - (i) Certain important sequences from 5' end of the DNA may be lost (ii) Important sequences from 3' end could be duplicated
 - (iii) Intron sequences cannot be studied (iv) Both (i) and (iii)
12. pMUTIN is a vector that facilitates integration of DNA fragment in _____ genome.
 - (i) *Bacillus* (ii) Yeast
 - (iii) *E. coli* (iv) Any of the above

13. _____promoter facilitates *Baculovirus* mediated expression in insect-cells.
- (i) *tac* (ii) T7
(iii) Gal1 (iv) Polyhedrin
14. Presence of _____ gene on the yeast cloning vector facilitates Red-White screening of recombinant clones.
- (i) URA3 (ii) *sup4*
(iii) Ade2 (iv) Aga2
15. Which of the following works as a viral vector for constructing transgenic plants?
- (i) Cauliflower mosaic virus (ii) Adeno-associated virus
(iii) *A. tumefaciens* (iv) Adenovirus
16. The limitation of *Agrobacterium* with respect to construction of transgenic plants is _____.
(i) It cannot efficiently applicable to dicots only (ii) It cannot modify nuclear genome
(iii) It cannot modify chloroplast genome (iv) Both (i) and (iii)
17. Use of intein as tag in fusion protein expression vectors facilitates_____.
- (i) soluble protein expression (ii) protein folding
(iii) protein export (iv) protein purification
18. Protein-ligand interactions can be studied using _____
- (i) yeast surface display (ii) yeast 2 hybrid system
(iii) yeast 3 hybrid system (iv) reverse 2 hybrid system
19. Which of the following is an ideal host for SV40 virus based vectors?
- (i) Simian cells (ii) Embryonic stem cells
(iii) Yeast cells (iv) Insect cells
20. _____ is a most widely explored selectable marker for animal cell transfection.
- (i) herpes simplex virus (HSV) Tk gene (ii) Adenovirus Tk gene
(iii) Gancyclovir (iv) Leu2 auxotrophic complementation

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Instructions:

- 1. Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I

Q - II Answer the following questions as directed 20

(A) What are plasmids? Explain various criteria to classify the plasmid vectors. 04

OR

(A) Discuss various DNA modifying enzymes used in rDNA technology 04

(B) Explain the following in brief (Any TWO) 06

- (i) Klenow fragment
- (ii) Dual selection strategy of pBR322
- (iii) Mechanism of iteron mediated regulation of plasmid copy numbers

(C) Give the features of Ti plasmid. Discuss an *Agrobacterium* based strategy to transfer genes in plant cells. 04

OR

(C) Discuss any two vectors for generation of transgenic animal cells. 04

(D) Write a short note on the following (Any TWO) 06

- (i) Direct methods of DNA transfer in animal cells
- (ii) Adenovirus and Adeno associated virus as vectors for animal cells
- (iii) Chloroplast transformation

SECTION – II

Q-III Answer the following questions as directed 30

(A) Explain the following (Any SIX) 18

- (i) Shuttle vectors and broad-host-range vectors
- (ii) Ideal features of an expression vector
- (iii) Hairpin loop strategy to construct cDNA
- (iv) Cosmids, plasmids and phages as cloning vectors
- (v) Spi selection phenotype for lambda phage vectors
- (vi) Advantages and disadvantages of M13 phage based vectors
- (vii) Disadvantages of *E. coli* as a host for recombinant protein expression
- (viii) Yeast surface display

(B) Answer the following (Any THREE) 12

- (i) Discuss strategies to avoid/overcome formation of inclusion bodies.
- (ii) Give an account on various types of cloning vectors for yeast host.
- (iii) Discuss strategies to screen genomic DNA library.
- (iv) Discuss pET vector system for optimizing gene expression in *E. coli*
- (v) Discuss a suitable strategy to promote export of heterologously expressed protein in *E. coli*.